AE410 Lab 2: Converging-Diverging Nozzle CFD for Varying Back Pressures

Lab Session: Monday, February 16, 2026
Due: Monday, February 23, 2026, 9am CST

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This laboratory assignment will focus on meshing, utilizing the converging-diverging nozzle geometry of Tutorial 2. Here we will conduct a grid convergence study, assess mesh quality, and utilize CFD data to study the influence of inlet pressure at a fixed back pressure.

Introduction

The Converging-Diverging (CD) Nozzle is a common nozzle in compressible flow studies that is also widely used in industry. It works uses converse properties of subsonic and supersonic flow to accelerate the air in both regimes. Subsonic air will accelerate in the converging section until the throat, where it turns supersonic and accelerates in the diverging section. For specific designs, the nozzle needs sufficiently high back pressures (p_0) to choke at the throat and not have a shock in the nozzle.

In this lab, we simulate the flow through a given CD nozzle at varying inlet pressures. The nozzle geometry is generated in Gmsh and exported in a format compatible with Su2. Both structured and unstructured meshes are used to examine the influence of mesh topology and refinement on solution accuracy. Simulations are performed for subsonic, shock-in-nozzle, and fully supersonic inlet conditions at a fixed back pressure.

A structured grid convergence study is conducted to find a good balance between mesh density and solver time. And a partitioned mesh is created to resolve the shock in the overexpanded nozzle case.

I. Grid Convergence Study (Supersonic Nozzle, Structured Mesh)

This section presents the results of the structured mesh grid convergence study for the supersonic nozzle case. The objective of this study is to evaluate the influence of mesh density on the predicted outlet mass flow rate and to assess solution consistency through comparison of Mach number contours across progressively refined grids. Four structured meshes were generated following the refinement relation $N = 2^n m + 1$, starting at $n = 0$ and SU2 was used to compute the steady compressible flow solution for each case.

A. Grid Convergence Study Plot

Figure 1 shows the variation of outlet mass flow rate with increasing mesh resolution. As the number of cells increases, the mass flow approaches an asymptotic value.

B. Best Mesh for Additional Simulations

The mesh resolution, total cell count, outlet mass flow rate, and approximate computational time on my personal computer for each structured grid are summarized in Table 1. As expected, increasing mesh resolution significantly increases computational cost and time while the predicted mass flow rate approaches an asymptotic value.

Table 1 Structured mesh refinement data for the supersonic nozzle case.

n	Mesh Size	Cell Count	$\dot{m}$ (kg/m-s)	Time (s)
0	31×51	1,794	-1419.751878	3.54
1	61×101	6,454	-1400.518595	12.02
2	121×201	24,772	-1395.915375	78.76
3	241×401	97,414	-1395.260684	384.52

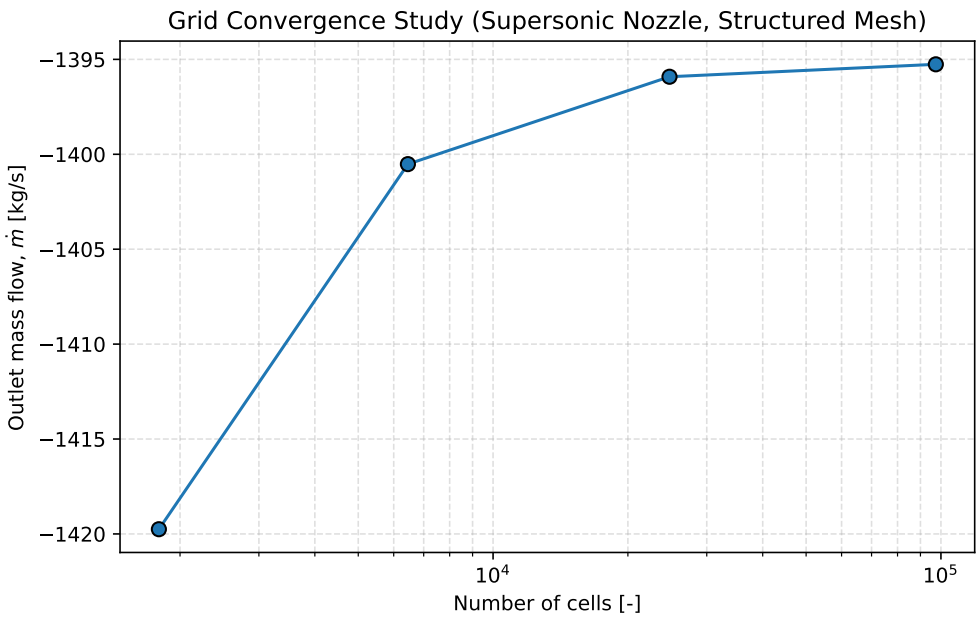


Fig. 1 Outlet mass flow rate as a function of total cell count for the structured meshes (logarithmic x-axis).

Based on the convergence behavior and computational cost, I would choose the $n = 2$ mesh for additional simulations. This mesh provides a mass flow rate sufficiently close to the value of the $n = 3$ case while not taking too long of a time to compute. The percent difference of $\dot{m}$ is only around 0.05% while the increase in time is 4.8 times. Also, a time of only a little over one minute is much easier to work around relative to human time compared to 6.5 minutes. Also, the $n = 2$ case is fine enough to capture the weak compression and expansion waves.

C. Mach Number Contours for All Structured Meshes

These contours are shown in figure 2.

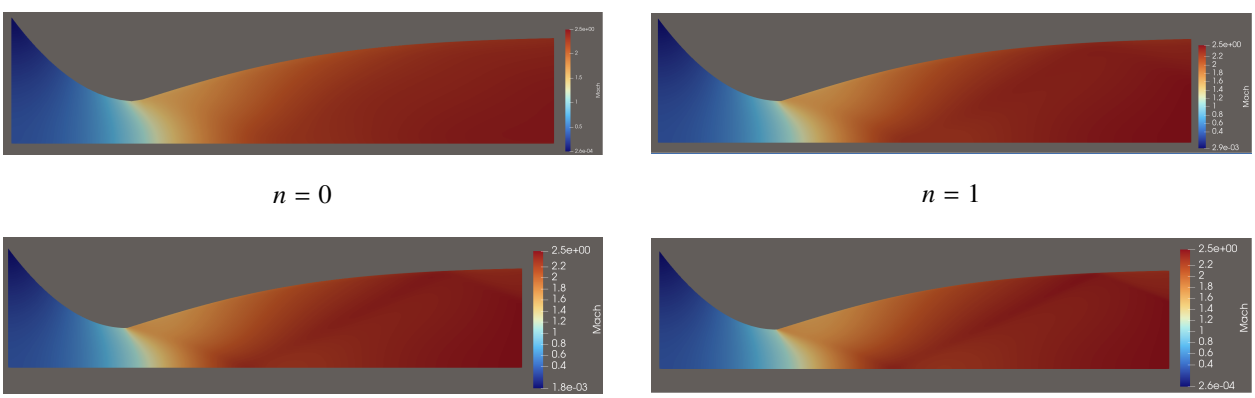


Fig. 2 Mach number contours for structured meshes with increasing refinement ($n = 0$ to $n = 3$).

II. Local Grid Refinement Process (Partitioned Mesh)

This section documents the mesh partitioning and local refinement strategy used to capture the normal shock in the shock-in-nozzle case. A partitioned mesh was generated to increase resolution in the vicinity of the shock while keeping the overall cell count and computational cost moderate.

A. Partitioned Mesh and Mach Number Contour

This is shown in figure 3.

B. Mesh Generation Strategy

To produce the partitioned mesh, an initial shock-in-nozzle solution was first run on a a fine unstructured mesh (lc = 0.05) to estimate the streamwise location of the normal shock . A mesh partition was then placed 0.5 meters upstream and downstream of this region so that I could create a structured mesh with a bump in the region. The reason I chose a hybrid mesh is because when trying to create the partition, I found it hard to resolve where the dense structured mesh and the sparser mesh intersected. An unstructured mesh would deal with that for me. When the partitioned mesh was first generated, the solver exhibited oscillatory behavior and failed to converge. After playing increasing the density of the transfinite curves, I realized that the innermost refined cells near the shock region had a very high aspect ratio, with elements that were significantly taller than they were wide. By reducing the number of nodes in the transfinite bump by a factor of two, the aspect ratio was at a more reasonable value. This led to the solution to converge more smoothly and with fewer iterations. This let me size the refined region such that the shock was resolved by multiple cells in the streamwise direction, minimizing visible contour artifacts and improving the sharpness of the pressure jump.

C. Centerline Mach Distribution and Shock Location

The shock location was identified from the centerline Mach distribution as the position of the steepest decrease in Mach number. Based on this criterion, the mean shock position was estimated to be $x \approx 2.759$ m.

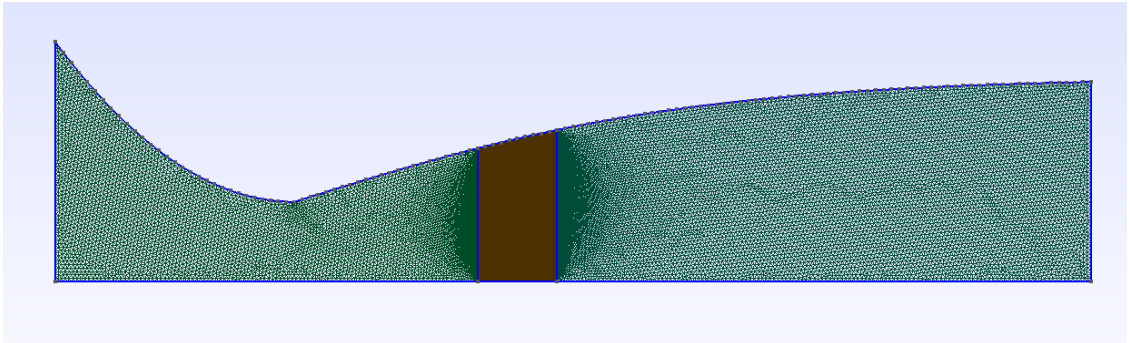
III. (Subsonic, Partitioned, and Supersonic Results

To compare the three operating conditions, the static pressure along the nozzle centerline ($y = 0$) was extracted from $x = -3$ m to $x = 10$ m and normalized by the corresponding inlet stagnation pressure, p_0, for each case. The three cases correspond to inlet pressures of 82 kPa (subsonic), 140 kPa (shock-in-nozzle), and 600 kPa (supersonic), with a fixed exit static pressure of 80.6 kPa.

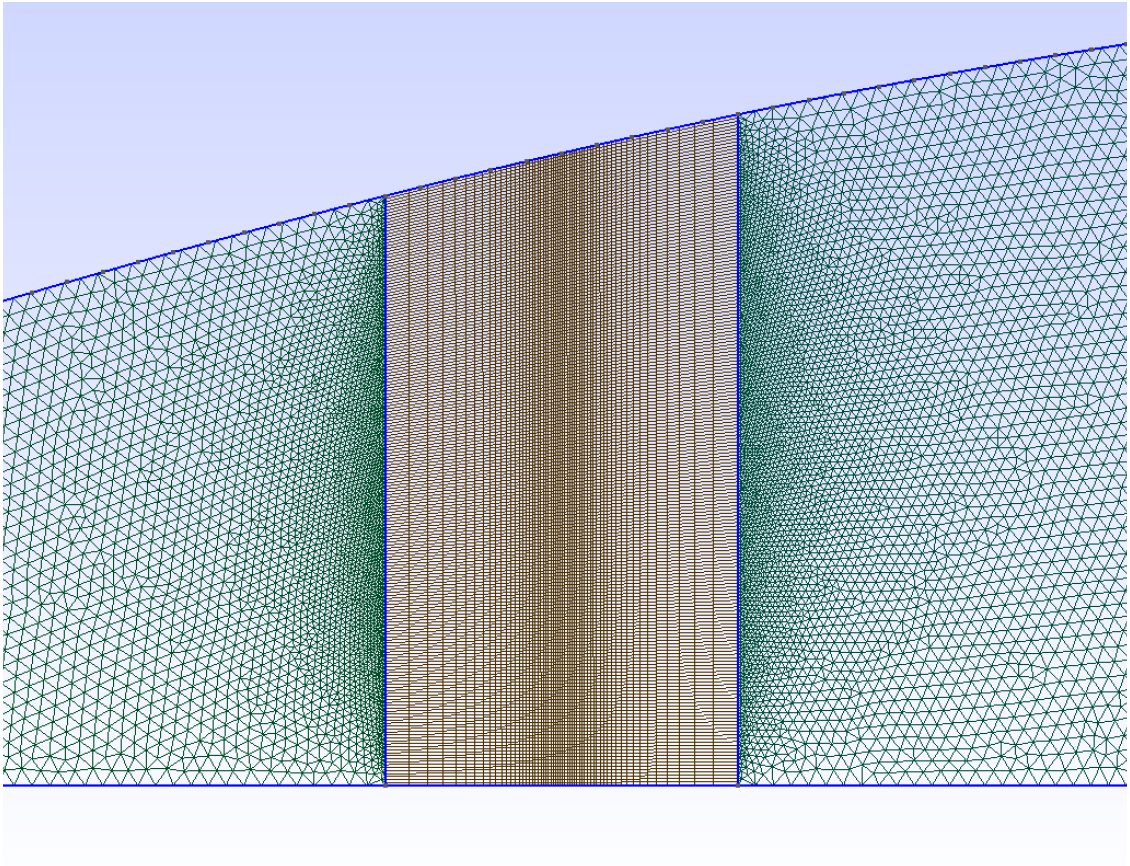
Figure 7 highlights the differing nozzle flow regimes across the three inlet pressure conditions. The subsonic case shows a smooth pressure variation through the nozzle, the shock-in-nozzle case exhibits a sharp pressure rise at the normal shock location, and the supersonic case maintains a lower pressure ratio through the diverging section consistent with strong expansion prior to reaching the imposed back pressure at the outlet.

Appendix

The github repo including the gmsh, Su2, and Paraview files as well as the python code used for post-processing, data analysis, and figure generation are available at: `https://github.com/nikitaaa27/AE462-git/tree/main/Lab2`



(a) Partitioned mesh showing the full computational domain.



(b) Close-up of the locally refined region near the shock location.

Fig. 3 Partitioned mesh used for the shock-in-nozzle case with local refinement around the normal shock region.

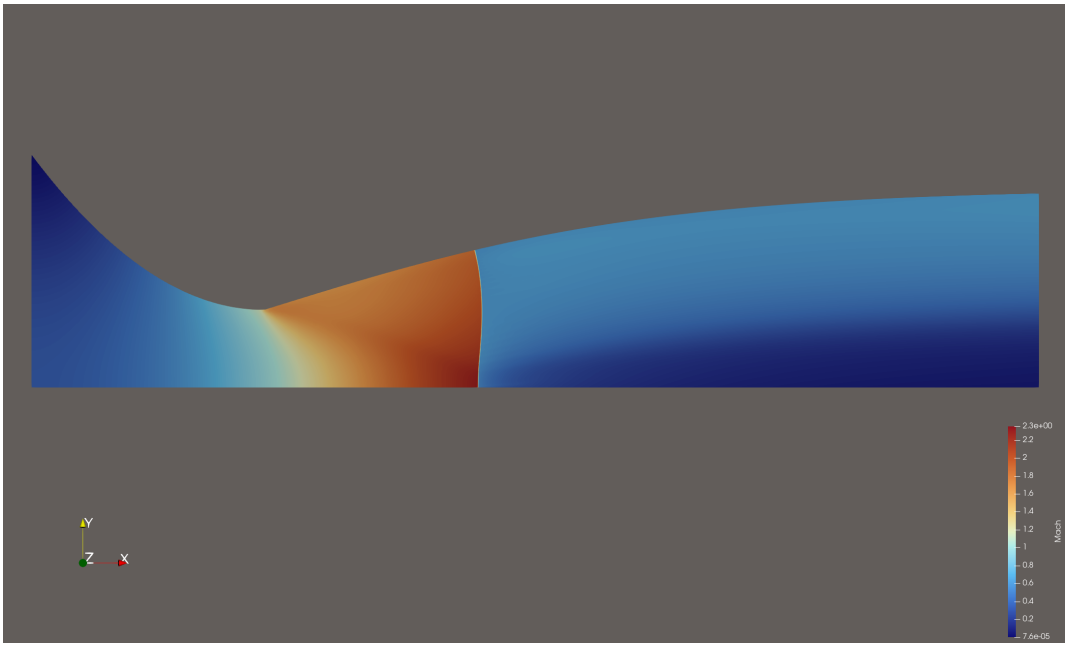


Fig. 4 Mach number contour for the shock-in-nozzle case using the partitioned mesh.

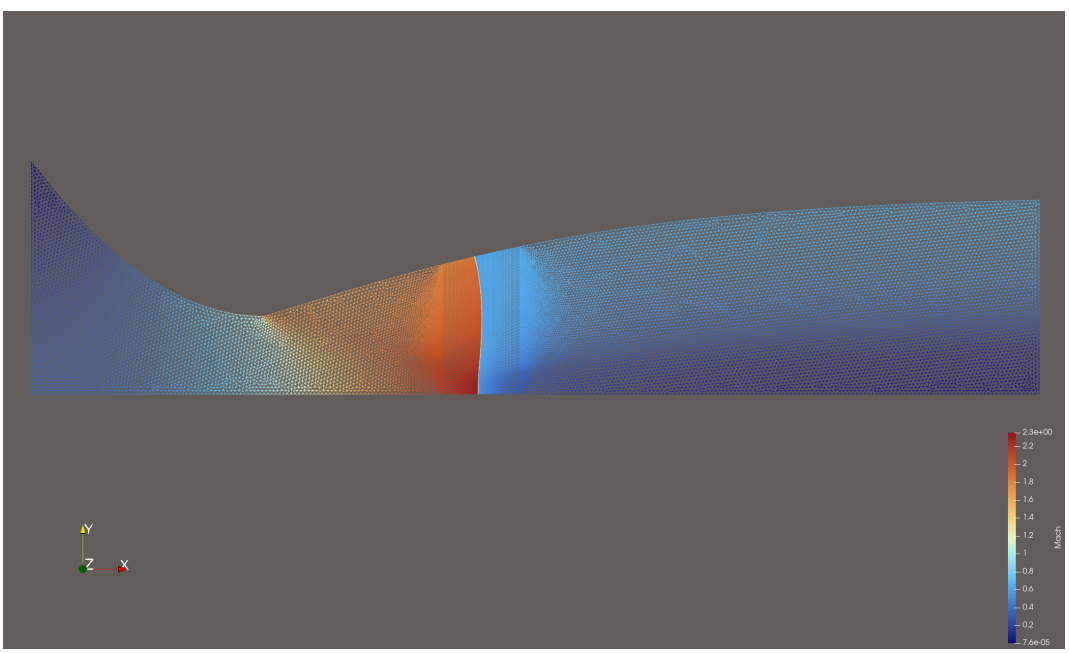


Fig. 5 Mach number contour with mesh wireframe overlay.

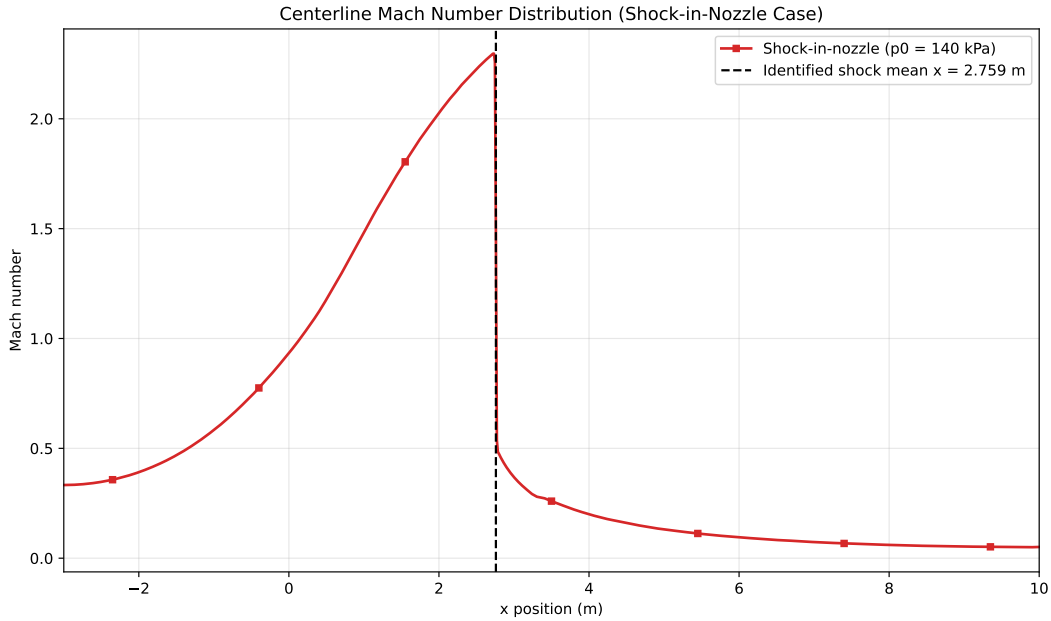


Fig. 6 Mach number distribution along the nozzle centerline for the shock-in-nozzle case. The shock location is identified by the sharp drop in Mach number.

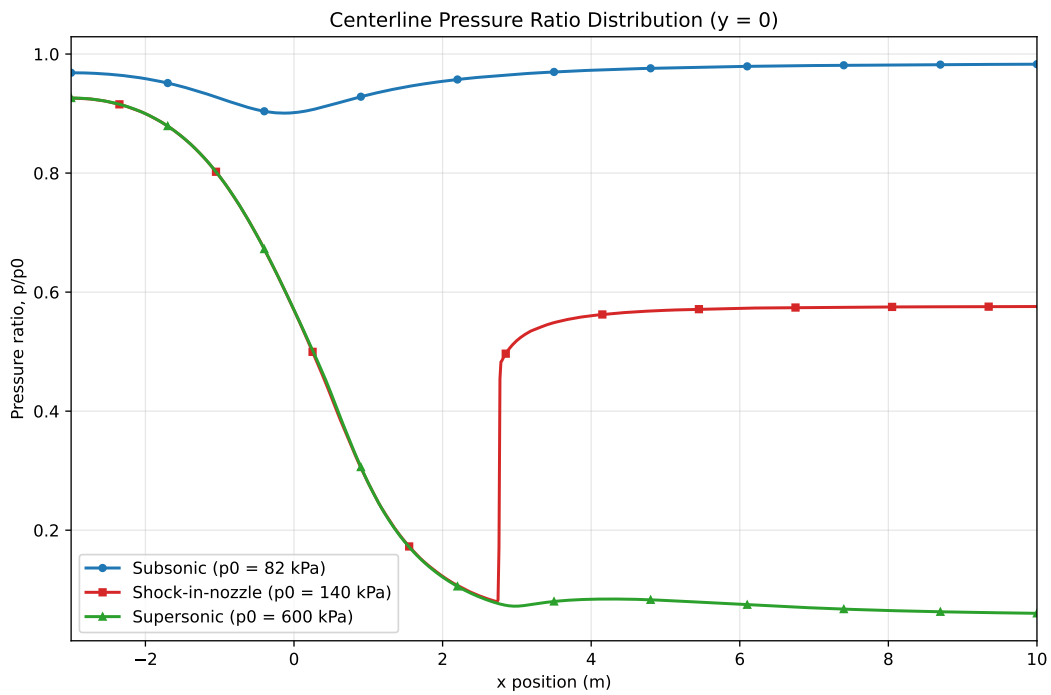


Fig. 7 Centerline pressure ratio distributions, p/p_0, along the nozzle for the three inlet pressure cases (subsonic, shock-in-nozzle, and supersonic).